

Physics Nobel Prize 2014 for LEDs as Future Lighting Devices



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According to Nobel committee citation, this year's Nobel Prize for Physics has been awarded to Isamu Akasaki and Hiroshi Amano, Nagoya University, Japan, and Shuji Nakamura, University of California, Santa Barbara, for the invention of efficient blue-light emitting diodes (LEDs). These have enabled bright and energy-saving white-light sources. In the spirit of Alfred Nobel, the prize rewards an invention of greatest benefit to mankind, and by using blue LEDs, white light can be created in a new way.

With the advent of LED lamps, we now have more long-lasting and efficient alternatives to older light sources. The committee added, "The LED lamp holds great promise for increasing the quality of life for over 1.5 billion people around the world who lack access to electricity grids. Due to low-power requirements, it can be powered using cheap, local solar power."

The blue LED invention has made possible a new energy-efficient and

environment-friendly light source. There are many different kinds of LEDs and most common types are categorised by colour, viewing angle, lens type, dimensions, forward voltage, forward current, packaging type and intensity.

Despite considerable efforts, both in the scientific community and in industry, the blue LED had remained a challenge for three decades. When these scientists produced bright-blue-light beams from their semiconductors in the early 1990s, they triggered a fundamental transformation of lighting technology. Red and green diodes had been around for a long time but, without blue light, white lamps could not be created. They succeeded where everyone else had failed.

The invention of blue LED is just twenty years old but, it has already contributed to create white light in an entirely new manner to benefit us all. White LED lamps emit a bright-white light, are long-lasting and energy-efficient. These are being constantly improved upon, getting more